

2.62 understand that platelets are involved in blood clotting, which prevents blood loss and the entry of micro-organisms

- 2.63 describe the structure of the heart and how it functions
- 2.64 explain how the heart rate changes during exercise and under the influence of adrenaline
- 2.65 describe the structure of arteries, veins and capillaries and understand their roles
- 2.66 understand the general structure of the circulation system to include the blood vessels to and from the heart, the lungs, the liver and the kidneys.

i) Excretion

Students will be assessed on their ability to:

Flowering plants

- 2.67 understand the origin of carbon dioxide and oxygen as waste products of metabolism and their loss from the stomata of a leaf

Humans

- 2.68 recall that the lungs, kidneys and skin are organs of excretion
- 2.69 understand how the kidney carries out its roles of excretion and osmoregulation
- 2.70 describe the structure of the urinary system, including the kidneys, ureters, bladder and urethra
- 2.71 describe the structure of a nephron, to include Bowman's capsule and glomerulus, convoluted tubules, loop of Henlé and collecting duct
- 2.72 describe ultrafiltration in the Bowman's capsule and the composition of the glomerular filtrate
- 2.73 understand that water is reabsorbed into the blood from the collecting duct
- 2.74 understand that selective reabsorption of glucose occurs at the proximal convoluted tubule
- 2.75 describe the role of ADH in regulating the water content of the blood
- 2.76 understand that urine contains water, urea and salts.

j) Coordination and response

Students will be assessed on their ability to:

- 2.77 understand that organisms are able to respond to changes in their environment
- 2.78 understand that homeostasis is the maintenance of a constant internal environment and that body water content and body temperature are both examples of homeostasis
- 2.79 understand that a coordinated response requires a stimulus, a receptor and an effector

Flowering plants

- 2.80 understand that plants respond to stimuli
- 2.81 describe the geotropic responses of roots and stems
- 2.82 describe positive phototropism of stems

Humans

- 2.83 describe how responses can be controlled by nervous or by hormonal communication and understand the differences between the two systems
- 2.84 understand that the central nervous system consists of the brain and spinal cord and is linked to sense organs by nerves
- 2.85 understand that stimulation of receptors in the sense organs sends electrical impulses along nerves into and out of the central nervous system, resulting in rapid responses
- 2.86 describe the structure and functioning of a simple reflex arc illustrated by the withdrawal of a finger from a hot object
- 2.87 describe the structure and function of the eye as a receptor